DRIFTLESS — Project Overview & Technical Objectives

1. Project Identity & Purpose

- **Project Name:** DRIFTLESS — Focus & Digital Wellbeing Companion
- **Repository Root:** `c:\Users\AADI\Desktop\driftless`
- **Application Class:** Hybrid Web Application & Browser Extension (Manifest V3 + Django REST Framework)
- **Primary Mission:** Empower knowledge workers, students, professionals, and digital creators to regain control of their digital attention by pairing real-time local web dwell tracking with server-side analytics, persona-based daily limits, proactive distraction escalation, and non-punitive focus streaks.

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2. Problem Statement

Modern computer users lose multiple hours each day to involuntary attention drifting—the subconscious habit of opening social media, video streaming, or infinite-scroll feeds while engaged in cognitive work. Existing screen-time software suffers from four core failures:

1. **Binary or Blurry Tracking:** Most trackers categorize domains into coarse "productive" or "unproductive" buckets without recognizing that an Instagram feed is distracting for an engineering student, yet essential research for a social media strategist or digital marketer.
2. **Passive vs. Actionable Enforcement:** Traditional tools display passive post-facto graphs after 4 hours of distraction have already passed, rather than intervening in real-time when a user reaches a threshold.
3. **Heavy Privacy Intrusion:** Conventional screen trackers log full page URLs, window titles, keystrokes, and tab screenshots, creating major data liabilities and personal privacy concerns.
4. **All-or-Nothing Blocking:** Hard website blockers trigger frustration and bypass habits. Users disable them within hours because they lack graceful escalation stages or emergency work sprints.

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3. Main Objectives

- **Zero-Friction Passive Dwell Tracking:** Measure exact active seconds spent on browser tabs using Chrome Manifest V3 APIs without background CPU churn, skipping background audio, idle states, or minimized windows.
- **Privacy-by-Design Architecture:** Restrict collected metrics strictly to bare domain names (`domain.com`) and accumulated time intervals (`seconds`), never saving paths, query strings, document titles, page content, or keystrokes.
- **Three-Tier Escalation Model:**
- *Tier 1 (80% Limit):* Non-intrusive gentle Chrome system notification nudge.
- *Tier 2 (100% Limit):* High-priority warning alert signaling that the user's daily streak is at risk, initiating a 5-minute countdown grace period.
- *Tier 3 (100% + 5m Grace Period):* Automatic dynamic network redirect via `declarativeNetRequest` to an uplifting, actionable `blocked.html` splash screen.
- **Persona-Driven Productivity Engine:** Support distinct operational presets (Student, Professional, Influencer, Creator, Other) that reconfigure default limits and reclassify domains (e.g., classifying `instagram.com` as Productive work for Influencers, while classifying `github.com` and `.edu` domains as Productive for Students).
- **Dual Visual Theme System:** Implement synchronized Light ("Calm Sky") and Dark ("Opal Noir") design tokens across the web dashboard, extension popup, and blocked screen without flash-of-unthemed-content (FOUC).
- **Self-Sovereign Data Controls:** Provide 1-click JSON full archive export and immediate irreversible browsing record purge endpoints.

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4. Intended Users & Personas

Persona	Primary Focus	Default Productive Sites	Typical Limits Applied
Student	Academic study, coding, homework	`.edu`, `github.com`, `leetcode.com`, `docs.google.com`, `notion.so`	Instagram: 30m, YouTube: 45m, Reddit: 20m, Netflix: 40m
Professional	Software engineering, collaboration, research	`github.com`, `stackoverflow.com`, `slack.com`, `jira.atlassian.com`, `linkedin.com`	Twitter/X: 20m, Instagram: 15m, YouTube: 30m, Reddit: 15m
Influencer	Content creation, community engagement	`instagram.com`, `youtube.com`, `tiktok.com`, `threads.net`, `canva.com`	Netflix: 30m, Twitch: 25m, Reddit: 20m

Persona	Primary Focus	Default Productive Sites	Typical Limits Applied
Creator	Video production, digital asset design, publishing	`figma.com`, `youtube.com`, `unsplash.com`, `notion.so`	Instagram: 45m, YouTube: 60m, Reddit: 20m, Netflix: 30m
Other / Custom	General balanced browsing	Default master map classification	Configurable via dashboard settings and extension popup

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5. Overall Application Architecture

```
graph TB
    subgraph ClientLayer["Client Layer (Chrome Browser)"]
        SW["Manifest V3 Service Worker<br/>(background.js)"]
        POPUP["Extension Popup Controller<br/>(popup.html / popup.js)"]
        BLOCK["DNR Block Screen<br/>(blocked.html)"]
        DASH_UI["Analytics Single-Page Dashboard<br/>(dashboard.html / Chart.js)"]
        AUTH_UI["Unified Auth & Persona Portal<br/>(auth.html)"]
    end

    subgraph StorageLayer["Browser Storage & Rules Engine"]
        LOCAL_STORAGE["chrome.storage.local<br/>(siteTimes, unsyncedDeltas, siteLimits,
token)"]
        DNR_ENGINE["declarativeNetRequest Engine<br/>(Dynamic Redirect Rules)"]
    end

    subgraph ServerLayer["Backend Application Layer (Django 4.2 LTS)"]
        URLS["Root URL Router<br/>(driftless/urls.py & core/urls.py)"]
        VIEWS["REST API Views & Controllers<br/>(core/views.py)"]
        SERIALIZERS["DRF Serializers & Validation<br/>(core/serializers.py)"]
        AUTH_BACKEND["Django Auth & Token Engine<br/>(TokenAuthentication)"]
    end

    subgraph DataLayer["Persistence Layer (SQLite Database)"]
        M_USER["User & UserProfile Models"]
        M_USAGE["UsageLog (Date, Domain, Seconds, Category)"]
        M_LIMITS["SiteLimit (Domain, Daily Limit, Enabled)"]
        M_SETTINGS["UserSettings (Goals, Streak, Modes)"]
        M_OVERRIDES["CategoryOverride & FocusSession"]
    end

    SW -- "Tracks active tabs & windows" ⟶ LOCAL_STORAGE
    SW -- "Installs dynamic block rules" ⟶ DNR_ENGINE
    POPUP -- "Reads local timers & controls mode" ⟶ LOCAL_STORAGE
    SW -- "POST /api/usage/ (Periodic Delta Sync)" ⟶ VIEWS
    SW -- "GET /api/limits/ & /api/settings/" ⟶ VIEWS
    DASH_UI -- "GET /api/summary/ (Analytics & Insights)" ⟶ VIEWS
    DASH_UI -- "PATCH /api/settings/ & /api/limits/" ⟶ VIEWS
    AUTH_UI -- "POST /api/auth/register/ & login/" ⟶ VIEWS

    VIEWS ⟶ SERIALIZERS
    SERIALIZERS ⟶ M_USER
    SERIALIZERS ⟶ M_USAGE
    SERIALIZERS ⟶ M_LIMITS
    SERIALIZERS ⟶ M_SETTINGS
    SERIALIZERS ⟶ M_OVERRIDES
```

6. Implementation Status Matrix

Major Component	Specific Feature	Implementation Status	Verification Status	Notes
Authentication	Username / Email / Phone Multi-Login	Fully Implemented	**PASS**	Validated via unit tests and live session generation.
Authentication	Persona Selection on Sign-Up	Fully Implemented	**PASS**	Auto-populates initial `SiteLimit` entries via `apply_persona_defaults`.
Authentication	DRF Token Regeneration	Fully Implemented	**PASS**	Invalidate old token and issues a fresh hex key.
Tracking Engine	Active Tab & Dwell Accumulation	Fully Implemented	**PASS**	Runs in `background.js`; pauses on window blur and idle state.
Tracking Engine	Idle Inactivity Filtering	Fully Implemented	**PASS**	`chrome.idle.setDetectionInterval(60)` suppresses idle time.
Escalation Engine	80% Nudge Notification	Fully Implemented	**PASS**	Triggers Chrome system notification on 80% limit reach.
Escalation Engine	100% Streak Alert	Fully Implemented	**PASS**	Warning notification flag sets `streakAtRisk=true`.
Escalation Engine	5m Grace Period & DNR Blocking	Fully Implemented	**PASS**	Dynamic DNR redirect rule points target domain to `blocked.html`.
Sync Engine	Periodic 60s Delta Sync	Fully Implemented	**PASS**	Pushes `unsyncedDeltas` array to `/api/usage/` via `chrome.alarms`.
Sync Engine	Midnight Local Reset	Fully Implemented	**PASS**	Flushes remaining time and resets daily local counters at 00:00.
Analytics Engine	Daily & 7-Day Usage Aggregation	Fully Implemented	**PASS**	Summarizes productive, social, entertainment, and other seconds.
Analytics Engine	Focus Efficiency Metric	Fully Implemented	**PASS**	Formula: `Productive / (Productive + Social + Entertainment) * 100`.
Analytics Engine	Strict Streak Calculation	Fully Implemented	**PASS**	Walks backward daily; breaks streak if any domain exceeded limit.
Focus Blocks	25, 45, 60m Focus Sprints	Fully Implemented	**PASS**	DNR rules block distracting sites; logs to `FocusSession`.
Theme System	Calm Sky & Opal Noir Dual Theme	Fully Implemented	**PASS**	CSS variables in `theme.css`, `data-theme` attribute, zero FOUC.
Data Privacy	JSON Data Export & Erasure	Fully Implemented	**PASS**	`/api/data/export/` and `/api/data/delete/` functional.

7. Known Limitations & Edge Cases

1. **Service Worker Suspension in Manifest V3:** Under Chrome's MV3 lifecycle rules, background service workers suspend after ~30 seconds of inactivity. Driftless mitigates this by committing timestamps on tab focus change, tab update, window blur, and firing `chrome.alarms` every 60 seconds to ensure timely wakeups.
2. **Tab Dwell Precision across Restarts:** If the browser process is killed abruptly via task manager without triggering `windows.onFocusChanged` or `tabs.onActivated`, time spent during that specific ongoing interval up to the last 60s flush may not be credited.
3. **Popup / Dashboard Independent Themes:** Both the extension popup and web dashboard respect the user's preferred theme key (`driftless-theme`), but since they execute in distinct DOM sandbox windows, changing the theme in one interface updates `localStorage` immediately, requiring a refresh or toggle click in the other.
4. **DeclarativeNetRequest Rule Limits:** Chrome enforces a quota of 5,000 dynamic rules per extension. Driftless consumes 1 rule per active focus block or exceeded site limit (typically < 20 rules), well within browser quotas.